ICAB June-July 2023 Exam
Case Study
Model Answer~~:~~

Draft Report

To : Mr. Masum Ahmed, Managing Director of PaperCraft Limited

From : Rashid Chowdhury FCA, Senior Partner, CRC

Date : 22 July 2023

Subject : Report on the ~~Assessment~~ assessment of business operation, advising on accounting & tax issues, regulatory matters, evaluation of strategic options, suggestion of investing and financing options, and analysis of ethical issues and sustainability factors of the business.

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1. Executive Summary:

1.1. Evaluation of financial and non-financial performances PCL.

Analysis of post accounting adjustment financial statement shows that financial performance of PCL was outstanding for the year 2023. It has achieved 21.1% growth in revenue which is much higher than 5% industry growth. Although its papers-based products achieved stunning 34.4% growth, ~~Significant~~ significant portion of revenue (58%) for the year has been contributed by Tissue & Hygiene products. Growth for Tissue & Hygiene products is seeming to be slowing down after Covid-19 pandemic is over, but it is still the largest cash generating unit for PCL.

PCL's cost of revenue for the year 2022 ~~is BDT 11,680 million which has~~ increased by 22.1% from last year due to increase of imported raw materials and imported duties. 60% of this increase has been caused by paper products. PCL gross profit ~~of BDT 2,929 million which~~ grew by 17.2% from prior year despite increases in imported raw materials and import duties. However, GP margin decreased slightly to 20% compared to 20.7% from previous year. PCL has achieved net profit margin of 3.1% which was 4.4% in the last year. Net profit margin has decreased by 16.4% ~~decreased~~ from prior year.

Current ratio ~~which~~ has decreased to 1.5 from 1.7 of previous year. Although liquidity position is good, but liquidity position has deteriorated from previous year. Current ratio is decreased due to significant increase in short term borrowings. ~~Its q~~Quick ratio of 0.3 also very poor which indicates that it can only pay 30% of its current liabilities with its quickly convertible current assets. PCL has 1.69 times of debt compared to its equity which has decreased form 2.44 of last year. Although PCL had excellent financial growth, it has not performed very well in terms of return of asset and equity.

PCL has performed excellently in terms of non-financial performance. It was able to diversify its business and introduce various paper based and alternative products. It has also maintained consisted revenue generation form export.

Recommendation to the board includes:

- Maintain current level of growth in revenue and decrease expenses.
- Restructure debt level to reduce financing cost.
- Increase focus on paper-based products which growing significantly
- Maintain sales contribution from tissue & hygiene products.
- Decrease inventory level to improve its working capital position
- Improve transaction recording, processing and presentation process.

1.2. SWOT Analysis.

PCL has built a strong brand reputation through quality and reliability which earned customer trust and a competitive edge over other competitors. It offers a wide range product makes it resilience against market fluctuations and changing consumer preferences. With its premium product quality, PCL was able to tap into international markets and increase its revenue from export.

PCL is heavily dependent on imported raw materials which could lead to disruption in business due to the ongoing dollar crisis. Alternatively, it can focus on environmentally sustainable source of raw materials like jute fibre to reduce reliance on expensive imports. PCL is committed in recycling of wastepaper and sourcing of sustainable raw materials which demonstrates its social responsibility. Furthermore, PCL boasts skilled and experienced workforce.

PCL is currently earning a significant sum from export revenue. It may further capitalize on its export success by identifying and penetrating new export destinations and leveraging its quality reputation.

Lack of financial oversight has led to weakness in financial reporting process and omission of financial transactions at the year-end reporting. This is a major weakness in PCL which need to be addressed. Furthermore, digitalization has led to reduction of paper consumption which could pose a long-term threat to PCL.

Recommendation to the board includes:

- PCL must strategically navigate the challenges posed by raw material dependency, digitalization, and competition.
- It should leverage its strengths in diversification, brand reputation, and environmental initiatives.
- PCL should seize opportunities in export of paper and paper products, exploring use of local raw materials including jute fibre in order to reduce its dependencies on imported raw materials.
- Management should focus on maintaining financial strength and year on year growth rate while improving its financial reporting process.

1.3. Analyse the Pros & Cons of strategic options:

Both strategic options have their pros and cons. Setting up Corrugated Cardboard Plant offers potential for growth and diversification but comes with a high initial investment and market-related risks. Acquisition of Papyrus Limited provides market entry and control over inputs but involves the uncertainty of acquisition costs and dependence on the pulp market. Both investment options will create additional income stream for PCL.

In case of Papyrus acquisition option, founding shareholders retaining 25% shares of Papyrus which can provide valuable expertise and support whereas new shares can extinguish interest-bearing liabilities improving Papyrus's financial position. However, integrating Papyrus into PCL's operations may present operational and cultural challenges for which PCL should made a detailed business plan.

Recommendation to the board includes:

- Investment option should be selected based on PCL's risk appetite, financial planning and long-term strategic goals, outcome of financial analysis and due diligence.
- Explore various financing options to fund the high initial investment. Consider a combination of equity and debt to optimize the capital structure.
- Review the financial calculations and assumptions used.
- Negotiate with papyrus management for favourable terms and conditions.

1.4. Recommendation and justification of investment options to be executed:

PCL board is currently considering between setting up a Corrugated Cardboard Plant (CCP) and acquiring 75% shares of Papyrus Limited. Both options are aligned with PCL's strategic goals. PCL apprises investment options using the Capital Asset Pricing Model (CAPM) which is determined to be 15%. Using the Required rate of return of 15%, the NPV analysis for setting up the Corrugated Cardboard Plant shows a positive NPV of 8,920 million Taka, indicating that this investment is financially attractive.

The NPV analysis for acquiring 75% shares of Papyrus reveals a negative NPV of 501 million Taka which suggest that this option would result in a financial loss. If PCL can negotiate a lower acquisition price for Papyrus, the NPV for this option might become positive and financially viable.

Recommendation to the board:

- PCL should invest in setting up the Corrugated Cardboard Plant (CCP) as this option not only generates a positive NPV but also diversifying PCL product line and adds a new income stream.
- Negotiate with Papyrus for reducing share price as the acquisition is still seen as strategically valuable. Favourable price can make NPV of Papyrus acquisition positive and financially viable.
- Perform due diligence on CCP prospective data and conduct a thorough risk assessment, develop mitigation plan to address any potential challenges that may arise during its implementation.

1.5. Calculation of EPS, Diluted EPS and effect of DEPS:

The analysis indicates that BSEC is likely to approve the proposed share issue at 50 Taka, including the 400% premium as PCL meets all the criteria in terms of NAV, EPS, and Market PE ratio. If PCL decides to issue shares at the proposed rate of 50 Taka per share, it will need to issue 52,980,000 new shares, resulting in a total of 505,054,701 shares. However, considering NAV per share, Sectoral PE and Average PE of comparable companies, PCL can issue shares at a price between 63 to 100 Taka.

At present PCL's basic EPS is determined to be 1.46 Taka. When it plans to issue new share, diluted EPS will be 1.31. PCL management should also consider other dilution impact on existing shareholders including ownership dilution, earnings & dividend dilution, voting rights dilution, and market price dilution.

Recommendation to the board:

- Proceed with share issuance through IPO as BSEC is likely to approve the proposed share issue at 50 Taka.
- PCL should consider issuing share at higher price considering the favourable market conditions, and higher NAV per share.
- Communicate with existing shareholders about the potential dilution impact of new shares issuance and ensure new share issuance are approved by board of directors or at AGM.
- PCL should also continue to evaluate alternative financing options, such as bank loans, in parallel to IPO financing option.

1.6. Feasibility bank loan financing and selecting the best option considering risks & benefits:

Financing the CCP project with a bank loan is a feasible option as the borrowing rate of 10% will result in higher NPV of the project compared to when evaluated using 15% CAMP rate. However, the acquisition of papyrus will not be financially feasible even though it is financed by bank loan.

Although the financing through bank loan raises risks of debt obligations, collateral requirements, interest costs, and credit risk, which can strain cash flow and pose financial risks, companies can be benefited from bank loan in terms of retaining control over company, tax deductibility, predictable repayment terms etc.

Risks of Financing Through IPO encompass loss of control, stringent disclosure requirements, market volatility, short-term pressure, and the costs associated with going public. But it also has some benefit in providing liquidity for founding investors, avoiding regular debt obligations, raising large capital quickly, and offering diversification opportunities to shareholders.

Recommendation to the board:

- Select CCP project and finance it through a bank loan, as it results in positive NPV when evaluated with required rate determined by PCL management and borrowing rate.
- Avoid acquisition of Papyrus and financing either through a bank loan or IPO.
- Understand the benefits of bank loan financing and ensure that the business can manage the associated risks like debt obligations, collateral requirements, interest costs, and credit risk.

1.7. Assessment of ethical issues in PCL's business and operation:

PCL's contribution to deforestation and the introduction of invasive species raises ethical questions about the responsible use of natural resources and the protection of local ecosystems and biodiversity. Reports of inadequate safety measures, exposure of workers to health risks, and potential child labor issues indicate a lack of ethical consideration for worker safety and labor practices.

The long-serving auditor, AIC, providing non-audit services and the lack of rotation of the engagement partner raise ethical concerns about auditor independence and potential conflicts of interest. The company's reluctance to change its auditor despite these issues further questions the company's ethical stance.

Recommendation to the board

- Adopt more sustainable and environmentally friendly practices, such as reducing water consumption, using alternative raw materials, and investing in energy-efficient machinery.
- Improve worker safety measures, conduct regular fire drills, and ensure compliance with labor laws.
- Remove AIC as auditor and appoint a new independent auditor following legal procedures.
- Maintain full financial transparency and comply with tax regulations and increase monitoring and transparency in recording and presenting financial transactions.

1.8. Assessment of sustainability factors of PCL's business:

The company is committed toward human sustainability but the delay in paying incentive bonus, use of child labor, worker exploitation, and chemical exposure raise significant concerns. PCL's economic sustainability is challenged by its import dependency, changing market trends, and competition. Moreover, borrowing may affect economic sustainability and financial stability.

PCL and the paper industry faces notable environmental sustainability challenges due to deforestation, waste generation, pollution, invasive species, and chemical exposure. PCL's economic sustainability can be enhanced by investments in technology and processes that reduce environmental impact.

Recommendation to the board

- Address concerns related to child labor, worker exploitation, and chemical exposure to ensure the safety and well-being of its workforce.
- Consider alternative, sustainable sources of raw materials to reduce the environmental impact and improve social sustainability.
- Introduce non-native tree species carefully to avoid potential environmental harm.
- Emphasize transparency and accountability, not only in financial reporting but also in addressing sustainability concerns.

2. Adjustment for reporting and compliance issues and evaluation of financial and non-financial performances of PCL. (Requirement-a-1)

2.1. Adjustments required for current reporting and compliance issues

Assessment of carrying value:

PCL purchased a machine in 2021 with 15 million which had expected useful life of 10 years. This means per year Depreciation is 1.5 million (15m/10). The asset has carrying value of 15m – (1.5m x 3) = 11.5 million. However, the asset has been damaged due to accident and requires regular maintenance. Fair value of the asset is only 7 million. As the asset is physically damaged, machine with new technology has emerged and realizable fair value of the asset is only 7 million, there is a clear indication of impairment. PCL should recognize an impairment loss of (11.5m – 7m) = 4.5 million (BDT 4,500,000) for the asset. Accordingly, PCL should post following adjustment entry:

Impairment loss on asset	Dr.	4.5 million	
Machineries	Cr.		4.5 million

VAT related issue:

VAT inspection has discovered that two years of VAT amounting to BDT 17.8 million has not been deposited. Pending amount of VAT has been deposited to govt. treasury but has not been accounted for. Furthermore, VAT authority is expected to penalize PCL by charging 50% of the pending VAT amount which is BDT 8.9 million Taka. Penalty amount is also not been accounted for in the draft financial statements. In order to account for the pending VAT amount and penalty, following entry need to be passed.

VAT Expenses	Dr.	17.8 million	
Penalty Expenses	Dr.	8.9 million	
Bank	Cr.		17.8 million
Provision for penalty	Cr.		8.9 million

It is to be noted that as pending VAT is not segregated year wise, it has been considered that prior year amount was not material and restatement is not required.

Foreign Exchange:

During the year PCL incurred interest on USD loan of USD 4,079,177 and recognized the entire amount as borrowing cost at BDT 93/$. That means total 4,079,177 x 93 = BDT 379,363,461 taka has been recognized as borrowing cost and has been capitalized. However, the average exchange rate was BDT 105/$ for the year which should be used as exchange rate as per IAS 21. Therefore, total interest expense for the year from USD loan should be (4,079,177 x 105) BDT 428,313,585. However, the entire amount should not be capitalized as building construction has completed 3 months before year end and another 3 months construction were ceased. Therefore, total 6-month interest (428,313,585 / 2) = 214,156,793 will be charged as interest expenses and remaining 6 months of interest (BDT 214,156,793) can be capitalized as part of construction cost. As construction cost has been overcharged, it should be reduced by (379,363,461-214,156,793) = 165,206,669. Differential amount will be adjusted with exchange loss recognized on outstanding loan balance. Therefore, following correction entry should be posted:

Finance Expenses	Dr.	214,156,793	
Construction in Progress	Cr.		165,206,669
Exchange Loss	Cr.		48,950,124

Product Supply Agreement:

On January 2023, PCL entered into an annual supply contract with QEL under which it needs to deliver 5,000 paper rim per month at the rate of 450 per rim. Failure to deliver the products will lead to a penalty of 450 x 10% = 45 Taka per rim undelivered. PCL has started to incur loss of BDT 15 per rim from April 2023. Since then, the contract become onerous as the penalty amount is higher compared to the loss due to delivery at lower price. Under this circumstance, PCL should recognize a loss immediately for reminder period of contract. As April to June loss has already been realized, PCL should recognize a loss for rest six month and create a contract liability. This contract liability should be adjusted with revenue over reminder of 6 month. Following entry should be passed to recognize the loss:

Loss on onerous contract	Dr.	450,000	
(5,000 rim x 6month x 15-taka loss per rim)			
Contract liabilities	Cr.		450,000

Inventory related issue:

There are 50,000 unsold paper rim which has been written down to realizable value of 280/rim in the prior year. These paper rims had cost of BDT 370/rim. However, at the end of current reporting period realizable value has increased to BDT 350/rim. As current year realizable value is lower than the cost value, inventory should be reported under current year realizable value instead of prior year realizable value. Therefore, Inventory value should be increased by 50,000 x 70 taka (350-280) = 3,500,000 = 3.5 million and cost of sales should be reduced by the same amount.

In order to correct the adjustment, following entry need to be posted:

Inventory	Dr.	3.5 million	
Cost of goods sold	Cr.		3.5 million

Unpaid incentive:

As the managing director has announced a incentive bonus for all of the employees, a valid expectation has been created for the employees. Therefore, PCL need to recognize a provision/ liability for the incentive payable equal to BDT 20,000,000. Therefore, following adjustment entry need to be posted:

Incentive expenses	Dr.	20 million	
Provision for incentive	Cr.		20 million

2.2. Evaluation of financial after adjustment

PCL's financial statement has been adjusted with the accounting adjustments mentioned above. Analysis has been performed on the adjusted financial statements.

Revenue:

Post accounting adjustment, PCL's revenue has not changed and stands at BDT 14,609,357,421 (14,609 million) which has massively increased by 21.1% (BDT 2,542 million) from prior year revenue of BDT 12,067,714,190. This increase is significantly higher compared to the industry growth rate of 5%. PCL's growth is mostly contributed by paper product which grew by 34% (1,551 million) from previous year and 61% of the total growth for the year. Rest 39% growth for the year has been contributed by tissue & hygiene products which grew by 13% from prior year. This is outstanding performance by PaperCraft Limited.

Of the total revenue, 42% has been contributed by paper products and 58% by the tissue & hygiene products. In the prior year revenue contribution was 38% from paper products and 62% from tissue & hygiene products. This clearly indicates that paper products have higher growth compared to the tissue & hygiene products. This might be due to increase in alternative use of paper-based products like paper cups, paper bags, paper cup, packing boxes. Furthermore, as COVID-19 pandemic is over, demand for tissue & hygiene products have decreased.

Almost 90% of the revenue has been generated from local sale and remaining revenue has been generated by export sales. This remains unchanged from prior year and expected to remain same in the future years. It is clearly evident that PCL is focusing on the local market only.

Despite negative impact of Covid-19 pandemics, scarcity of raw materials and digitalization in the office work, PCL managed to increase its revenue at much higher rate than the industry growth rate. This is outstanding performance by PCL.

Cost of Revenue:

PCL's cost of revenue for the year 2022 has been determined at BDT 11,680,399,628 (11,680 million) which has increased by 22.1% (BDT 2,112 million) from cost of revenue of BDT 9,568,034,401 for 2021. Cost of revenue has increased at a higher rate compared to the revenue growth rate. This is increased due to increase of imported raw materials and imported duties. Although Tissue & Hygiene products contributes 60% of the cost of revenue but increase of cost of revenue is due to 13% growth in cost of paper products. Growth of cost of revenue for paper products contributed 60% of the growth in cost of revenue for the year.

GP and GP Margin:

Adjusted gross profit for the year risen to BDT 2,929 million which is 17.2% higher (BDT 429 million) from prior year gross profit of BDT 2,500 million. This is excellent performance by PCL compared to the industry growth and despite increase in imported raw materials and import duties.

During the year PCL GP margin has slightly decreased from 20% compared to 20.7% that of prior year. This decrease in GP margin is due to higher growth rate of cost of sale compared to revenue growth rate. This is in line with market scenarios where material cost has increased. Despite that GP margin has not significantly deteriorated which is very good performance by PCL.

GP margin has been contributed equally from both paper-based products and tissue & hygiene products. Paper based products has GP margin of 23% which decreased from 27% from previous year. On the other hand, tissue & hygiene products have GP margin of 17.3% which has increased 16.9% that of previous year.

Net Profit Margin

PCL's pre-adjusted net profit for the year was BDT 660,202,581 which was 24% higher than prior year. However, due to net increasing adjustment of BDT 213,356,669 in the expenses, adjusted profit decreased to BDT 446,845,912 which is 16.4% lower than the prior year net profit. This decrease is almost entirely due to massive increasing adjustments in the finance cost for foreign currency denominated loan. Other than the adjustment, net profit generated by PCL is impressive considering the industry growth and foreign currency volatility. However, adjusted net profit margin is only 3% which is very nominal considering the business size.

Current and Quick Ratio

Calculation of post adjustment financial statement of PCL shows that its current ratio for the year ended 30 June 2023 is 1.5 which has decreased slightly from current ratio of 1.7 in 2023. The decrease in current ratio is due to decrease in current asset for payment of VAT and increase of Trade and other payables due to penalty provision. Because of this adjustment current asset of 2023 has decreased and current liability has increased. Accordingly, the current asset ratio has decreased during the year.

Current ratio of PCL for both 2023 and 2022 is below 2 which is considered to be the standard. However, the current ratio is still over 1 which indicates PCL has sufficient current assets to meet its current liabilities. A decreased in current ratio indicates the necessity of investing in the working capital. PCL has already invested its capital with short term borrowings but still the working capital has been decreasing. This might be due to devaluation of Taka against foreign currency which has increased the foreign currency denominated loans.

PCL's quick ration for the year 2023 is only 0.3 which was 0.4 in the year 2022. PCL quick ratio was already very low in the prior year, and it has got even lower in the current year. This shows a very poor financial performance by PCL. Quick ratio shows how quickly an entity can liquidate its assets to pay off the current obligations. Analysis of the quick ratio shows that PCL most of the current asset are not quickly convertible to cash and only 30% obligation can be paid off by liquidated immediately its current assets. This is due to significant in the current liabilities and very large volume of inventories in the current assets.

Debt to Equity and TIE Ratio

PCL is highly geared company, but its gearing has decreased in the current year. Its gearing ratio in 2023 is 1.69 which was 2.44 in prior year. This is significantly decrease in gearing position. It appears PCL has been trying to decrease its gearing level in the capital structure. It has almost decreased its gearing to half. However, the decrease actually due to increase in revaluation reserve. That is why the finance cost have not decreased compared to the prior year rather it has increased by 25% from the comparative year. Because of this reason, one may challenge that revaluation of asset has been done to increase the equity component which will eventually lead to decrease in the Debt-to-equity ratio and bring it to an acceptable level.

Although the finance cost has increased significantly compared to the prior year, Times Interest Earned (TIE) ratio has not changed. TIE ratio remains stable at 1.4 times in both current and prior year. This indicates PCL has been able to generate profit 1.4 times of the finance cost. This may be acceptable level in the short run but in the long run the TIE ratio is not sufficient as sudden decrease in profitability will lead to liquidity crisis and PCL might not be able to service its debts.

Asset Turnover and Return on Assets

Despite a massive increase in revenue and profitability, asset turnover is very poor for PCL. Its current asset turnover is 0.32 which also in the prior year. This could be caused by significant investment in the property, plant & equipment. PCL is not efficient enough to generate sufficient revenue utilizing the resource at hand. Compared to the investment in fixed asset, revenue of the company should been higher. However, this might be industry norm as well which management need to review.

PCL's return on asset is 1% in 2023 which has decreased from 1.4% in 2022. Return on asset has decreased due to decrease in profitability arising from adjustments in the finance cost. As PCL has substantial amount of foreign currency denominated loan, it is only natural that it would have finance cost would be higher.

Return on Equity

PCL return on equity has decreased to 2.7% in 2023 compared to 4.9% of 2022. Despite increase in revenue, net profit for the year have decreased. Furthermore, due to increase in revaluation reserve, total equity amount has increase. Therefore, overall return on equity has decreased. As ROE has almost halved. PCL was not efficient enough to create wealth for the shareholders.

2.3. Evaluation of non-financial performance

PCL is currently producing various products which includes paper-based products and tissue & hygiene products. PCL has well diversified its products. Its product range includes white writing & printing papers, newsprint, duplex board, liner paper, Kraft paper, Carbonless Paper, Coloured/ Offset Printing & Wrapping Papers, Colour Ledger Paper, Sticker Papers, Cigarette Tipping Papers, OGR Paper, Exercise Books, A4 Paper, Paper Sack, Art card, art paper, allied products, facial tissue, toilet paper, paper towel, Kitchen napkin. It has production capacity of 30,000 Metric Ton Paper per year which is huge production capacity compared to other companies in the industry where factories are closing.

There is a high potential for paper industry to export. PCL has been exporting its finished products to various country which ranges from 10% to 15%. In the current year, 10% of total revenue has been generated from export sales.

Despite the industry is facing problems with availability of raw materials, PCL has been able to ensure supply of raw materials. It is also considering opening its own pulp production company which will ensure stable supply for the PCL and other companies.

Overall PCL has shown outstanding non-financial performance over the years.

2.4. Conclusion

- Financial performance of PCL was outstanding for the year 2023. Its has achieved 21.1% growth in revenue. It papers-based products achieved stunning 34.4% growth from previous year. These growths are way higher compared to the industry growth of 5%. One of the reasons for revenue growth is scarcity of paper product in the market due to shortage of raw material.

- Despite Tissue & Hygiene products contributes more than 58% in the revenue, it has only grown 13% during the year and contributed 39% in total revenue growth. Growth for Tissue & Hygiene products is seeming to be slowing down after Covid-19 pandemic is over. However, this line of products are still the largest cash generating unit for PCL.

- PCL's cost of revenue for the year 2022 is BDT 11,680 million which has increased by 22.1% from last year due to increase of imported raw materials and imported duties. Growth in cost of revenue is higher compared to revenue growth rate. Growth of cost of revenue for paper products contributed 60% of the growth in cost of revenue for the year.

- PCL had an excellent performance in generating gross profit of BDT 2,929 million which grew by 17.2% from prior year. This is higher than industry growth despite increases in imported raw materials and import duties. However, PCL GP margin decreased slightly to 20% compared to 20.7% from previous year. GP margin has been contributed equally from both paper-based products and tissue & hygiene products.

- PCL has achieved net profit margin of 3.1% which was 4.4% in the last year. Net profit margin has decreased by 16.4% decreased from prior year. Net profit of PCL was already very low comparing the gross profit earned, but its net profit has further decreased due to increase in operating expenses including significant adjustments in finance cost.

- Current ratio which has decreased to 1.5 from 1.7 of previous year. This indicates that a good liquidity position and have ability to pay off its current obligation. But liquidity position has deteriorated from previous year. Current ratio is decreased due to significant increase in short term borrowings. Its quick ratio of 0.3 also very poor which indicates that it can only pay 30% of its current liabilities with its quickly convertible current assets.

- PCL is highly geared entity as it has 1.69 times of debt compared to its equity. However, its debt to equity has decreased from 2.44 indicates that PCL was able to decrease gearing in the current year. This might be the reason why quick ratio of the company is very low.

- Although PCL had excellent financial growth, it has not performed very well in terms of return of asset and equity.

- PCL has performed excellently in terms of non-financial performance. It was able to diversify its business and introduce various paper based and alternative products. It has also maintained consisted revenue generation form export.

2.5. Recommendation

- PCL should aim to continue holding or increasing current level of growth in revenue. But it should aim to reduce its expenses in order to increase overall profitability.
- Of the total revenue generated, significant portion has been used up in servicing debt. PCL should try to restructure its debt to reduce finance cost.

- PCL should focus on paper-based products has it has been growing significantly while maintaining tissue & hygiene product sales which contribute more than 50% of revenue.

- PCL should try to decrease inventory level to improve its quick ratio. Furthermore, it should improve asset utilization as the asset turnover is very poor.

- Management should properly incorporate all the necessary adjustment entry at the year end.

3. Evaluate of the Strengths, Weaknesses, Opportunities, and Threats (SWOT) of PCL's existing business (Requirement a-ii):

3.1. SWOT Analysis:

Strengths

- ***Diversified Product Portfolio:***
  PCL offers a wide range of paper and hygiene products, including tissues and other diversified items. This diversification provides resilience against market fluctuations and changing consumer preferences.

- ***Strong Brand Reputation:***
  PCL has cultivated a strong brand reputation in the industry, known for quality and reliability. This reputation gives the company a competitive edge and customer trust.

- ***Export Growth:***
  The significant increase in paper and paper product exports, especially to the US, Eurozone, Middle East, and Africa, highlights PCL's ability to tap into international markets and adapt to global demand.

- ***Environmental Initiatives:***
  PCL's commitment to recycling wastepaper aligns with growing environmental awareness and stricter regulations. This not only demonstrates social responsibility but also provides a sustainable source of raw materials.

- ***Skilled Workforce:***
  PCL boasts a skilled and experienced workforce capable of operating its manufacturing units efficiently, ensuring product quality and operational excellence.

Weakness

- ***Raw Material Dependency:***
  The company's heavy reliance on imported raw materials has become a vulnerability due to the ongoing dollar crisis, leading to increased production costs and supply chain disruptions.

- ***Quality and Technology Gap:***
  Outdated machinery and the use of inferior-quality raw materials in some plants can result in production inefficiencies, reduced product quality, and higher costs. This could potentially lead to going concern issue for PCL.

- ***Incorrect financial reporting:***
  Several financial transactions have been missed out or incorrectly reported during the year end financial statements which shows there are weakness in financial reporting oversight roles.

- ***Decreasing female employee gender ratio:***
  Over the year gender ratio of female employees has been declining as the female employees are leaving their jobs. This might be due to unfavourable working condition for female employees.

Opportunities

- ***Focus on Tissue and Hygiene Products:***
  The growing demand for tissue and hygiene products, especially amid the COVID-19 pandemic, presents a significant opportunity for PCL to expand its presence in these segments.

- ***Recycling Wastepaper:***
  The shortage of dollars for importing raw materials creates an opportunity for PCL to invest in expanding its wastepaper recycling capabilities, reducing dependency on imports and enhancing sustainability.

- ***Local Raw Materials:***
  Exploring the use of jute fiber and other locally available raw materials can reduce reliance on expensive imports and contribute to environmental conservation.

- ***Export Expansion:***
  PCL can further capitalize on its export success by identifying and penetrating new export destinations, diversifying its customer base, and leveraging its quality reputation.

- ***Investment in Technology:***
  Modernizing production facilities with state-of-the-art technology can improve efficiency, reduce costs, and enhance product quality, addressing some of the weaknesses.

- ***Research on Jute Fibre:***
  The research on using jute fibre as a raw material for pulp and paper production presents an opportunity for PCL to explore a locally sourced, eco-friendly alternative to imported raw materials.

Threats

- ***Dollar Crisis and Import Challenges:***
  The persistent dollar crisis and import restrictions on raw materials pose a substantial threat to PCL's production costs, profitability, and supply chain stability.

- ***Digitalization and Declining Demand for Printing Paper:***
  The ongoing shift toward digital media and reduced paper consumption in offices and education institutions pose a long-term threat to traditional paper product demand.

- ***Intensified Competition:***
  The competitive landscape in the paper and hygiene industry remains intense, with established and emerging players vying for market share, potentially leading to price wars and margin pressures. PCL faces stiff competition both locally and globally, requiring consistent innovation, quality improvement, and cost-efficiency to maintain market share and profitability.

- ***Rising Raw Material Costs:***
  The soaring prices of paper due to raw material shortages can erode profit margins, making it challenging to maintain competitive pricing.

- ***Regulatory Changes:***
  Evolving environmental regulations may necessitate costly compliance measures, potentially impacting PCL's cost structure and profitability.

- ***Market Uncertainties:***
  Global economic and geopolitical uncertainties can impact export markets, currency exchange rates, and overall market demand, affecting PCL's international expansion plans.

- ***Environmental Regulations:***
  Stricter environmental regulations may necessitate further investments in cleaner production processes, waste management, and sustainability initiatives, potentially increasing operational costs.

3.2. Conclusion:

- PCL offers a wide range product makes it resilience against market fluctuations and changing consumer preferences.
- PCL has built a strong brand reputation through quality and reliability which earned customer trust and a competitive edge over other competitors.
- PCL was able to tap into international markets and increase its revenue from export of paper and paper product.
- PCL's commitment to recycling wastepaper and sourcing of sustainable raw materials demonstrates its social responsibility.
- PCL ensures product quality and operational excellence by boasting skilled and experienced workforce.
- PCL is heavily dependent on imported raw materials which could lead to disruption in business due to the ongoing dollar crisis.
- Omission of financial transactions at the year-end reporting indicates weakness in financial reporting process and lack of financial reporting oversight.
- Use of jute fibre can reduce reliance on expensive imports and become and environmentally sustainable source of raw materials.
- PCL may further capitalize on its export success by identifying and penetrating new export destinations and leveraging its quality reputation.
- Government may introduce new laws and regulations imposing restriction and monitoring to the e-commerce sites and online marketplaces.
- Digitalization has led to reduction of paper consumption which could pose a long-term threat to PCL.

3.3. Recommendation:

- PaperCraft Limited must strategically navigate the challenges posed by raw material dependency, digitalization, and competition.
- It should leverage its strengths in diversification, brand reputation, and environmental initiatives.
- PCL should seize opportunities in the tissue and hygiene segments, exploring use of local raw materials including jute fibre in order to reduce its dependencies on imported raw materials.

- PCL should make prudent investments in technology which can position the company for long-term success.
- PCL should also aim for mitigating threats related to the dollar crisis and regulatory changes by focusing on sustainability factors and compliances.
- PCL should explore opportunities of exporting paper and paper products to other countries.
- PCL should take action to improve its financial reporting process in order to avoid omission of financial transactions.
- Management should focus on maintaining financial strength and year on year growth rate.
- Comply with government laws and regulations in order to avoid regulatory and compliance risk.

4. Analyse the Pros & Cons of strategic options (Requirement B-i):

PCL's directors are currently considering between two investment options. First option is setting up a new plant solely for corrugated cardboard and other paper-based packaging materials. and second option is to acquire 75% shares of Papyrus Limited (Papyrus) which is engaged in production of pulps made of cellulose fibres. Pros and Cons of each option is assessed below:

4.1. Setting up Corrugated Cardboard Plant (CCP):

Pros	Cons
1. ***Potential for Growth:*** The demand for packaging materials is rising, indicating a viable business opportunity for PCL. 2. ***Control:*** PCL will have full control over the operations of the corrugated cardboard plant. 3. ***Diversification:*** This option allows PCL to diversify its business by entering a new market segment. 4. ***Diversification of business:*** Setting up corrugated cardboard plant will enable diversification of products and reduce business risks. 5. ***Predictable Cash Flows:*** Cash flows are forecasted for the next 10 years, providing clarity on expected returns.	1. ***High Initial Investment:*** The project requires a significant initial investment of 5,000 million BDT. 2. ***Risk of Market Saturation:*** Over time, the market for corrugated cardboard may become saturated, impacting profitability. 3. ***Operational Risk:*** Running a new plant involves operational challenges and risks. 4. ***Perpetual Growth Assumption:*** The perpetuity growth rate assumption of 3% may not hold in the long term, affecting the accuracy of projections.

4.2. Acquisition of Papyrus Limited:

Pros	Cons
1. ***Market Entry:*** Acquiring Papyrus allows PCL to enter a new market segment related to pulp production. 2. ***Control Over Inputs:*** PCL gains control over a critical input (pulp) for its paper mills, ensuring a stable supply. 3. ***Readily Available Market:*** Papyrus already has an established market and sourcing network. 4. ***Partial Ownership:*** Founding shareholders retaining 25% shares can provide valuable expertise and support. 5. ***Use of New Shares for Debt Reduction:*** The use of new shares to extinguish interest-bearing liabilities can improve Papyrus's financial position.	1. ***Acquisition Cost:*** The cost of acquiring 75% shares in Papyrus is not specified, and it could be substantial. 2. ***Integration Challenges:*** Integrating Papyrus into PCL's operations may present operational and cultural challenges. 3. ***Financial Performance:*** Papyrus has posted a net loss in the provided financial statements, which may impact profitability. 4. ***Dependency on Pulp Market:*** PCL's success in this investment is tied to the performance of the pulp market, which can be volatile. 5. ***Assumed Growth Rate:*** The perpetuity growth rate assumption of 4% may not be realized, affecting future cash flows.

4.3. Conclusion:

- Setting up Corrugated Cardboard Plant offers potential for growth and diversification but comes with a high initial investment and market-related risks.
- Acquisition of Papyrus Limited provides market entry and control over inputs but involves the uncertainty of acquisition costs and dependence on the pulp market.
- Investment in either option will create additional income stream for PCL.
- Founding shareholders retaining 25% shares of Papyrus which can provide valuable expertise and support whereas new shares can extinguish interest-bearing liabilities improving Papyrus's financial position.
- Integrating Papyrus into PCL's operations may present operational and cultural challenges.
- Papyrus already has an established market and sourcing network of which PCL can take benefit. Similarly, setting up CCP will enable PCL to tap into new market of packaging materials.
- Each acquisition will lead to increase control over PCL’s value chain.

4.4. Recommendation:

- The strategic options should be chosen considering PCL's risk tolerance, financial capacity, and long-term strategic goals.
- Perform a thorough financial analysis and due diligence before making a decision.
- Before proceeding with available options, conduct a comprehensive market analysis to assess the long-term viability of the corrugated cardboard business, taking into account potential market saturation. and management capabilities.
- Explore various financing options to fund the high initial investment. Consider a combination of equity and debt to optimize the capital structure.
- Review the financial calculations and assumptions used.
- Develop a risk mitigation strategy to address operational challenges and uncertainties associated with running a new plant. This could involve setting up effective quality control processes and supply chain management.
- Assess how the acquisition of Papyrus aligns with PCL's diversification goals and its ability to control critical input supplies for its paper mills.
- During the acquisition negotiation, seek favourable terms and conditions that benefit PCL, including agreements on control, management, and the use of new shares for debt reduction before finalizing the acquisition deal.

5. Recommendation and justification of investment options to be executed (Requirement b-ii):

PCL board is currently considering between two investment options. First option is to set up a Corrugated Cardboard Plant and second option is to acquire 75% shares of Papyrus. Setting up CCP will add increase product width and diversify the business as CCP will produce packaging materials which is different from PCL's regular production. On the other hand, Papyrus is engaged in pulp manufacturing. By acquiring 75% shares of Papyrus, PCL will be able to ensure constant supply of pulp as raw materials for its paper production plants. PCL and the entire paper industry is having difficulties with import of expensive raw materials and high import duties. As papyrus is focusing on production of locally made pulp, PCL will be highly benefited and able to increase its revenue by fulfilling local and international demand though increased production.

Both investment options require significant cash outflow but adds new income stream for PCL. Both strategic options are fit for PCL strategic goals. However, ultimate investment decision will be taken based on return on investment and Net Present Value (NPV) analysis.

PCL board apprises the investment options using Capital Asset pricing Model (CAPM). From the information provided Risk Free return is 7.5%, Market Return is 12.5% and market beta is 1.5. Considering is CAPM rate is calculated at 7.5%+ 1.5x (12.5%-7.5%) = 7.5% + 7.5% = 15%. NPV of both options are analysed below using the CAPM rate.

5.1. NPV of CCP:

Investment in CCP will require BDT 5,000 million. Management has estimated following free cash flow for next 10 years. After the 10th year, cash flows are expected to grow at 3% perpetually. NPV of the CCP project is calculated below:

Year	Free Cash Flow (A)	Terminal Value ^ (B)	Discount Rate * (C)	Discounted Cash Flow D = (A+B)*C	Net present value E= Sum(D)
0	(5,000)	-	1.0000	(5,000)	8,920
1	430	-	0.8696	374	
2	835	-	0.7561	631	
3	1,110	-	0.6575	730	
4	1,366	-	0.5718	781	
5	1,652	-	0.4972	821	
6	1,945	-	0.4323	841	
7	2,230	-	0.3759	838	
8	2,505	-	0.3269	819	
9	2,790	-	0.2843	793	
10	3,078	26,420	0.2472	7,292	

^ Terminal value = FCF x (1+g)/(d-g) where d is discount rate and g is growth rate.

* From present value chart for $uPVIF_{15\%,10}$

Based on the calculation, it is expected that setting up CCP will generate NPV of 8,920 million. As NPV of the CCP project is positive, it may be accepted.

5.2. NPV of acquisition of Papyrus:

PCL board is currently planning to acquire 75% shares of Papyrus Limited. At present papyrus has net asset of 883 million out of which 600 million is share capital with 60 million issued shares (Face value of 10 Taka). As per agreement with PCL management, Papyrus management will retain 25% ownership and issue new shares for 75% shares that PCL will acquire. That means post acquisition, Papyrus share capital will be 600M/0.25 = 2,400M taka which means total number of shares will be 240 million. Therefore, new shares to be issued to PCL is 240 – 60 = 180 million. If PCL acquires share at the rate of Net Asset per share, PCL will have pay 180M x (883/60) = 2,649 million Taka.

PCL management has prepared a post-acquisition projected cash flows from 2024 to 2028 which is expected to grow thereafter at 4% perpetually. Cash flows will be discounted at 15% as required by PCL management. Furthermore, as the capital structure consist of 100% equity, there will not be any adjustment in the discounting rate. Based on expected acquisition cost and future cash flow of the business, NPV of aquation option is calculated below:

	2024	2025	2026	2027	2028	Enterprise value (In million)	Share of PCL (75%)	Payment for Shares	NPV
FCF	611	247	212	246	359	2,864	2,148	(2,649)	(501)
Terminal value	-	-	-	-	3,394				
Discount factor	0.8696	0.7561	0.6575	0.5718	0.4972				
Present Value	531	187	139	141	1,866				

Based on the calculation, it is expected that acquisition of 75% shares of Papyrus will generate negative NPV of 501 million. As NPV of the project is negative, it should be discarded. However, if PCL can reduce the acquisition price by negotiate, NPV might become positive.

5.3. Suggestion on investment option with justification:

PCL board is currently considering between two options: setting up CCP and acquisition of Papyrus. Both of the options are strategically fit to PCL and will add value to the business. However, financial analysis and NPV analysis shows that investment in CCP will generate positive NPV of 8,920 million and acquisition of Papyrus will generate negative NPV of 501. On this basis it will be appropriate for PCL to invest on setting up Corrugated Cardboard Plant. It will not only increase product line but also add new income stream.

5.4. Conclusion:

- PCL board is currently considering between investment options of setting up a Corrugated Cardboard Plant (CCP) and acquiring 75% shares of Papyrus Limited.
- Both options are aligned with PCL's strategic goals. CCP would diversify the business by producing packaging materials, while acquiring Papyrus would ensure a constant supply of pulp, addressing raw material supply issues.
- PCL apprises investment options using the Capital Asset Pricing Model (CAPM) which is determined to be 15%.
- Using the Required rate of return of 15%, the NPV analysis for setting up the Corrugated Cardboard Plant shows a positive NPV of 8,920 million Taka, indicating that this investment is financially attractive.
- The NPV analysis for acquiring 75% shares of Papyrus Limited reveals a negative NPV of 501 million Taka which suggest that this option would result in a financial loss.
- If PCL can negotiate a lower acquisition price for Papyrus, the NPV for this option might become positive and financially viable.

5.5. Recommendation:

- Based on the NPV analysis, it is advisable for PCL to invest in setting up the Corrugated Cardboard Plant (CCP). This option not only generates a positive NPV but also aligns with PCL's strategic goals by diversifying its product line and adding a new income stream.
- PCL should try to negotiate with Papyrus Limited for reducing acquisition share price as the acquisition is still seen as strategically valuable. If a more favorable acquisition price can be secured, the NPV for the Papyrus option may become positive and financially viable.
- PCL should reconsider Papyrus Acquisition as the analysis indicates that acquiring 75% shares of Papyrus Limited would result in a negative NPV of 501 million Taka.
- PCL should actively monitor the market conditions, especially regarding raw material prices and availability, to ensuring that the CCP project remains financially sound.
- Although CCP is financially viable, PCL should perform due diligence on prospective financial data and conduct a thorough risk assessment, develop mitigation plan to address any potential challenges that may arise during its implementation.

6. Calculation of EPS, Diluted EPS and effect of DEPS (Requirement b-iii):

PCL management is considering financing the strategic options either through bank loan or through issuing new shares through IPO. If management plans to acquire Papyrus through issuance of share through IPO, PCL need to raise at least BDT 2,649 million after issuance cost. PCL is planning to issue shares at 400% premium (i.e. Face Value 10 Taka + Premium of 40 Taka = 50 Taka per share) subject to approval of Bangladesh Securities & Exchange Commission (BSEC).

6.1. Determining acceptability of proposed premium and number of shares to be issued:

BSEC consider the NAV, EPS, and sectoral Market PE ratio, etc. while approving the premium rate. Therefore, PCL needs to know whether the 400% premium might be acceptable to the BSEC.

PCL has currently 452,074,701 issued shares with face value of BDT 10 and according to the latest financial statements, its total asset of BDT 45,310 million. Therefore, its NAV per share is (45,310 million/ 452,074,701) = 100.22 Taka.

According to the latest financial statements, PCL net profit is BDT 660,202,581. Therefore, Earnings Per Share (EPS) is (660,202,581/452,074,701) = 1.46.

On 30 June 2023, Sectoral PE of paper industry at Dhaka Stock Exchange was 51.98. Based on this PE, share price of PCL can be valued at 51.98 x 1.46 = 75.89 Taka.

Average Share price of the comparable companies in the same industry was (94 +350 +266 +820)/4 = 382.5 Taka whereas Average EPS was (2.98 + 5.80 + 4.20 + 22.50)/4 = 8.87. Therefore, PE ratio of comparable companies were 382.5/8.87 =43.12. Considering average PE of comparable companies, PCL share price can be valued at BDT 43.12 x 1.46 = 62.96 ~ 63 Taka.

From the analysis it can be shown that PCL can fulfil BSEC criteria in terms of NAV, EPS and Market PE ratio. Therefore, BSEC is likely to approve PCL proposed share issue at 50 Taka including 400% premium. In fact, PCL may issue at higher price for example at 70 Taka considering marker PE and higher NAV than share price.

If PCL plans to issue share at existing proposed rate of BDT 50 taka per share, it needs to issue (2,649 million / 50) =52,980,000 new shares. In that case its total number shares will become 452,074,701 + 52,980,000 = 505,054,701 shares.

6.2. Calculation of EPS and DEPS

As per earlier calculation, PCL has Basic EPS of (660,202,581/452,074,701) = 1.46.
Considering prospective shares to be issued, PCL's diluted EPS will be (660,202,581/505,054,701) = 1.31.

6.3. Dilution impact on the existing shareholders

When a company issues new shares, it can have several effects on existing shareholders. The dilution impact on existing shareholders when new shares are issued includes:

- ***Ownership Dilution:***
  When a company issues new shares, the ownership percentage of existing shareholders decreases proportionally.

- ***Earnings and Dividend Dilution:***
  Dilution also affects existing shareholders' earnings and dividend per share. When new shares are issued, the company's earnings are distributed over a larger number of shares, potentially reducing earnings per share (EPS) and, consequently, dividend per share (DPS).

- ***Voting Rights Dilution:***
  If the new shares are issued, existing shareholders may see their relative voting power reduced. This can impact their ability to influence corporate decisions and governance.

- ***Market Price Dilution:***
  The issuance of new shares can lead to a decrease in the market price of the company's stock. This is because the market may perceive the increase in the number of shares as a signal of potential dilution of value. Lower stock prices can impact the overall value of existing shareholders' holdings.

6.4. Conclusion:

- PCL's NAV per share is calculated to be 100.22 Taka, which is well above the face value of 10 Taka.
- Based on the paper industry's sectoral PE ratio, PCL's share price is estimated at 75.89 Taka. Additionally, the average PE ratio of comparable companies suggests a share price of 63 Taka.
- The analysis indicates that BSEC is likely to approve the proposed share issue at 50 Taka, including the 400% premium as PCL meets all the criteria in terms of NAV, EPS, and Market PE ratio.
- If PCL decides to issue shares at the proposed rate of 50 Taka per share, it will need to issue 52,980,000 new shares, resulting in a total of 505,054,701 shares.
- PCL's basic EPS is determined to be 1.46 Taka and diluted EPS will be 1.31 when new shares will be issued through IPO when acquiring papyrus.
- When issuing new shares through IPO, PCL management should consider the dilution impact on existing shareholders including ownership dilution, earnings & dividend dilution, voting rights dilution, and market price dilution.

6.5. Recommendation:

- Proceed with share issuance through IPO as BSEC is likely to approve the proposed share issue at 50 Taka, including the 400% premium as PCL meets all required criteria.
- PCL should consider issuing share at higher price considering the favourable market conditions, and higher NAV per share. However, this should be carefully evaluated to avoid discouraging potential investors.
- PCL should provide clear information and actively communicate with existing shareholders about the potential dilution impact of new shares issuance on ownership percentages, earnings per share, dividend per share, voting rights, and the market price of the stock.

- PCL should establish a comprehensive investor relations strategy to maintain transparency and build trust with existing and potential investors.
- PCL should also continue to evaluate alternative financing options, such as bank loans, in parallel to IPO financing option.
- Monitor market conditions and investor sentiment which can change rapidly. It is important to be flexible in adjusting the timing and pricing of the IPO to maximize its success.
- Engage legal and financial advisors with expertise to ensure compliance with all regulatory requirements and to make informed decisions regarding the IPO.
- Ensure that the issuance of new shares through an IPO is approved by the company's board of directors and existing shareholders to avoid any legal or governance issues down the road.

7. Feasibility of financing options through bank loan and determining the best financing option considering risks and benefits (Requirement b-iv):

7.1. Feasibility of Financing through bank loan:

As financing strategy, PCL management is also considering feasibility of financing the project through bank loan. Bank is proving loan at Six Month Moving Average Rate of Treasury Bill (SMART) rate plus 3% corridor. Current SMART is 7% which means current borrowing rate will be 7% + 3% = 10%. This borrowing rate is much lower than the CAMP rate of 15% require by PCL management.

Financing CCP with bank loan:

CCP already has positive NPV after discounting the Free Cash Flows at 15%. If the project is financed by bank loan, cost of capital will become 10% which is much lower than the 15% CAMP rate. Therefore, NPV of the project will be even higher. This means financing CCP with bank loan is feasible.

Financing Papyrus acquisition with bank loan:

Acquisition of Papyrus generates a negative NPV when the cash flows are discounted at 15% CAPM rate. But if it is financed by bank loan cost of capital will be at 10% and accordingly revised NPV will be calculated as:

	2024	**2025**	**2026**	**2027**	**2028**
Post-Acquisition FCF by Papyrus	611	247	212	246	359
PCL %	75%	75%	75%	75%	75%
FCF controlled by PCL	458.25	185.25	159.00	184.50	269.25
Terminal value (3,394 x 75%)	-	-	-	-	2,545.64
Discount factor	0.9091	0.8264	0.7513	0.6830	0.6209
Present Value	417	153	119	126	1,748

PCL Share of Papyrus enterprise value : 2,563 million
Investment made : (2,649) million
NPV : (86) million.

Revised calculation with 10% discount rate on acquisition of papyrus shares shows a negative NPV of 86 million. That means even of the acquisition is financed through bank loan, it will not be feasible.

7.2. Risk & benefits of financing through bank loan:

Risks of financing through bank loan:

1. ***Debt Obligations****:*
   Businesses must make regular interest and principal payments, which can strain cash flow, especially if the business is not performing well.

2. ***Collateral Requirements:***
   Many bank loans require collateral, such as assets or personal guarantees, which can be risky if the business fails to repay the loan.

3. ***Interest Costs:***
   Over time, the interest paid on a bank loan can add up, potentially making it a more expensive form of financing than equity.

4. ***Credit Risk:***
   Businesses with lower credit ratings may face higher interest rates and difficulty securing loans.

***Benefits of financing through bank loan*:**

1. ***Control:***
   Borrowers retain full control of their company since they don't have to share ownership with external shareholders. The lender has no say in the company's operations.

2. ***Interest Tax Deductibility:***
   The interest paid on a business loan is typically tax-deductible, which can reduce the overall cost of borrowing.

3. ***Predictable Repayment:***
   Loan terms are usually structured with fixed monthly payments, making it easier for businesses to budget and plan for repayments.

4. ***Faster Process:***
   Securing a bank loan is quicker than going through the IPO process which is lengthy and complex.

5. ***No Dilution:***
   Taking out a loan does not dilute existing shareholders' ownership in the company.

7.3. Risk & benefit of financing through IPO:

***Risks of financing through IPO*:**

1. **Loss of Control**:
   Going public means sharing ownership and control with a broader group of shareholders, potentially leading to conflicts over the company's direction.

2. **Disclosure Requirements**:
   Publicly traded companies must adhere to stringent regulatory and reporting requirements, which can be costly and time-consuming.

3. **Market Volatility**:
   Stock prices can be highly volatile, impacting shareholder value and making it challenging for long-term planning.

4. **Short-Term Pressure**:
   Public companies may face pressure to deliver short-term results to satisfy shareholders, potentially sacrificing long-term strategic goals.

5. **Costs of Going Public**:
   The IPO process itself can be expensive due to underwriting fees, legal expenses, and compliance costs.

***Benefits of financing through IPO*:**

1. **Large Capital Injection**:
   An IPO can raise a substantial amount of capital quickly, allowing for significant business expansion, debt reduction, or investment in new projects.

2. **Liquidity for Founders and Early Investors**:
   Founders and early investors can sell some of their shares, providing liquidity and realizing their investments.

3. **Enhanced Visibility**:
   Going public can enhance a company's visibility and credibility, making it easier to attract customers, partners, and employees.

4. **No Debt Obligations**:
   Unlike loans, equity financing does not require regular interest or principal payments, reducing financial strain.

5. **Diversification**:
   Shareholders can benefit from diversification by holding a portfolio of different stocks.

7.4. Conclusion:

- Financing the CCP project with a bank loan is a feasible option as the borrowing rate of 10% will result in higher NPV of the project compared to when evaluated using 15% CAMP rate.
- Financing the acquisition of Papyrus through a bank loan does not appear to be feasible even with the lower cost of capital (10%) as the revised NPV still shows a negative value of -86 million.
- Financing through bank loan raises risks of debt obligations, collateral requirements, interest costs, and credit risk, which can strain cash flow and pose financial risks.
- Benefits of bank loan financing include retaining control over company, tax deductibility, predictable repayment terms, faster financing compared to IPO, and no dilution effect.
- Risks of Financing Through IPO encompass loss of control, stringent disclosure requirements, market volatility, short-term pressure, and the costs associated with going public.
- Benefits of IPO financing include, providing liquidity for founding investors, avoiding regular debt obligations, raising large capital quickly, and offering diversification opportunities to shareholders.

7.5. Recommendation:

- Select CCP project and finance it through a bank loan, as it results in positive NPV when evaluated with required rate determined by PCL management and borrowing rate.
- Avoid acquisition of Papyrus and financing either through a bank loan or IPO.
- Consider the benefits of bank loan financing and ensure that the business can manage the risks associated with bank loans like debt obligations, collateral requirements, interest costs, and credit risk.
- Understand the risks of going public, including the loss of control, stringent disclosure requirements, market volatility, short-term pressure, and the costs involved in the IPO process.
- Carefully weigh the advantages and disadvantages of each option and consider the long-term impact on the business's financial stability and ownership structure.

8. Assessment of ethical issues in PCL's business and operation (Requirement C-i).

8.1. Ethical issues:

Ethical issues involving PCL's business and operation includes:

Environmental Impact:
The paper industry is known for its high-water consumption, deforestation, and pollution. PCL's operations have significant environmental impacts. Its reliance on wood pulp, release of chlorinated organic compounds, and generation of substantial solid waste, including sludge from wastewater treatment plants, contribute to environmental degradation. Additionally, the use of dioxins in the paper production process and the lack of energy-efficient machinery further deteriorates these issues.

Resource Depletion:
PCL's consumption of trees for paper production contributes to deforestation and the depletion of natural resources. The introduction of invasive species, such as *Acacia auriculaeformis* and *Acacia mangium*, poses a threat to local ecosystems and biodiversity. This raises ethical concerns about the responsible use of natural resources.

Worker Safety and Labor Practices:
Reports of fire incidents in paper mills, inadequate safety equipment, and expired fire extinguishers suggest a lack of concern for worker safety. Moreover, forcing workers to handle toxic chemicals without proper safety gear, exposing them to health risks, and employing child labor during peak seasons are serious ethical violations.

False Advertising:
If PCL claims to use recycled materials in its products but does not actually do so, it engages in false advertising. This can mislead consumers who prioritize environmentally friendly products and violate principles of honesty and transparency.

Financial and Tax Issues:
The company's failure to account for VAT and penalty amounts in its financial statements, as well as the potential understatement of the carrying value of assets (e.g., the damaged machine), may raise concerns about financial transparency and tax compliance.

Appointment of auditor:

AIC has been auditor of the company who has been serving for last 10 years and providing various non-audit service like internal audit, tax planning & calculation, filling of statutory and tax returns. Performing non-audit service while appointed as auditor clearly violates auditor independence. Furthermore, Engagement partner has not been rotated in last years which may create familiarity threat. PCL chairman has referred AIC to other companies which may be the financial reason for AIC not changing its PCL audit fee in last four years. PCL management, despite knowing all these issues, is not planning to change its auditor. This could raise question on ethical standpoint of PCL management.

8.2. Conclusion

- PCL's contribution to deforestation and the introduction of invasive species raises ethical questions about the responsible use of natural resources and the protection of local ecosystems and biodiversity.
- Reports of inadequate safety measures, exposure of workers to health risks, and potential child labor issues indicate a lack of ethical consideration for worker safety and labor practices.
- Concerns about financial transparency and tax compliance arise from PCL's failure to account for VAT and penalty amounts in its financial statements and the potential understatement of asset values. These issues raise ethical questions about the company's financial practices.
- The long-serving auditor, AIC, providing non-audit services and the lack of rotation of the engagement partner raise ethical concerns about auditor independence and potential conflicts of interest. The company's reluctance to change its auditor despite these issues further questions the company's ethical stance.

8.3. Recommendation

- Adopt more sustainable and environmentally friendly practices, such as reducing water consumption, using alternative raw materials, and investing in energy-efficient machinery.
- Improve worker safety measures, conduct regular fire drills, and ensure compliance with labor laws.
- Be transparent in advertising and marketing materials, accurately representing the use of recycled materials in products.
- Maintain full financial transparency and comply with tax regulations.
- Remove AIC as auditor and appoint a new independent auditor following legal procedures.
- Increase monitoring and transparency in recording and presenting financial transactions.

9. Assessment of sustainability factors of PCL's business (Requirement C-ii).

All business needs to be sustainable in order to continue its business for the foreseeable future. A business can earn sustainability when it achieves Human Sustainability, Social Sustainability, Economic Sustainability and Environmental Sustainability.

9.1. Human Sustainability:

Human sustainability focuses on maintaining and improving the human capital in society by investments in the health and education systems, access to services, nutrition, knowledge and skills. Following human sustainability factors has been identified from PCL business operation.

Employee Welfare:
The company mentions its commitment to equal employment opportunities, employing a significant number of female workers, and plans to improve gender balance in the future. It also discusses recruiting from nearby localities to boost local employment opportunities.

Occupational Health and Safety:
The company has developed comprehensive policies and procedures related to occupational health, safety, and welfare of its employees and associated persons. It aims to ensure the health and safety of its employees and others who may be exposed to risks.

Unpaid Incentive:
The company's announcement of a 10% incentive bonus for employees is a potential human sustainability factor, but the fact that it has not been paid yet raises questions about the company's commitment to employee welfare.

Child Labor and Worker Exploitation:
The information suggests concerns about child labor, forced overtime during peak seasons, and workers being exposed to toxic chemicals without safety equipment. These issues are significant human sustainability concerns.

Safety Concerns:
Fire incidents in paper mills, lack of safety equipment, and faulty electric wiring indicate a lack of safety measures, posing risks to workers' safety and well-being.

9.2. Social Sustainability:

Social Sustainability is focused on identifying and managing business impacts, both positive and negative, on people. Social sustainability aims to preserve social capital by investing and creating services to communities and its cultures. Following social sustainability factors has been identified from PCL business operation.

CSR Initiatives:
The company's CSR activities, including support for education, medical camps, and sponsorship of sports tournaments, contribute to social sustainability by addressing social needs and improving the well-being of the community.

Environmental Impact:
While primarily an environmental factor, the environmental impact of the paper industry, including deforestation, waste generation, and air and water pollution, also has social sustainability implications by affecting the overall quality of life and health of communities.

Recycling Practices:
The practice of falsely claiming the use of recycled materials in products can be a social sustainability concern, as it can mislead consumers and undermine efforts to promote sustainable practices.

Use of Invasive Species:
The introduction of invasive tree species for pulp production can have negative social sustainability effects by disrupting ecosystems and potentially impacting local communities and indigenous populations.

9.3. Economic Sustainability:

Economic sustainability aims to maintain the financial capital and improve the standard of living. In the context of business, it refers to the efficient use of assets to maintain company profitability over time. Following economic sustainability factors has been identified from PCL business operation.

Import Dependency:
PCL's reliance on imported raw materials, especially pulp, and its efforts to find alternative sources impact its economic sustainability due to potential cost increases and supply chain disruptions.

Market Trends:
Adapting to changing market trends, such as the shift towards digital communication and the decrease in demand for certain paper products, is essential for long-term economic sustainability.

Competitiveness:
PCL's ability to compete with plastic products and adapt to consumer preferences for environmentally friendly alternatives impacts its economic sustainability.

Investment and Technology:
Investing in modern technology, improving pulp production, and implementing energy-efficient processes can enhance economic sustainability while reducing environmental impact.

Financial Reporting and Compliance:
Maintaining compliance with International Financial Reporting Standards (IFRSs) and addressing issues related to financial reporting, such as unpaid incentives, is crucial for economic sustainability and maintaining stakeholders' trust.

Foreign Borrowing:
PCL's strategy to focus on foreign borrowing due to increasing local borrowing costs can impact its economic sustainability and financial stability.

9.4. Environmental Sustainability:

Environmental sustainability aims to improve human welfare through the protection of natural capital (e.g., land, air, water, minerals etc.). Following environmental sustainability factors has been identified from PCL business operation.

Deforestation and Resource Usage:
The paper industry contributes to deforestation and high resource consumption, including wood and water. It mentions that 35% of cut wood is used for paper production, and one ton of paper consumes 90 tons of water.

Waste Generation:
The industry generates significant waste, including paper waste, sludge, and solid waste, causing environmental problems. It's noted that paper waste accounts for 40% of global waste.

Pollution:
Paper production processes release pollutants such as dioxins into the environment, which are highly toxic and harmful to human health. It also mentions emissions of various air pollutants and gases.

Invasive Species:
The introduction of non-native tree species like Acacia and eucalyptus for pulp production is recognized as an environmental issue, as these species can become invasive and harm local ecosystems.

Fire Incidents:
Frequent fire incidents in paper mills are mentioned, indicating potential safety and environmental risks associated with inadequate fire safety measures.

Chemical Exposure:
Workers in paper mills are exposed to toxic chemicals without proper safety equipment, which can pose health and environmental risks.

9.5. Conclusion

- The company is committed toward human sustainability but the delay in paying incentive bonus, use of child labor, worker exploitation, and chemical exposure raise significant concerns.
- The company has engaged in CSR activities, which contribute positively to social sustainability. But it also has negative impacts on society due to deforestation and pollution.
- PCL's economic sustainability is challenged by its import dependency, changing market trends, and competition. Moreover, borrowing may affect economic sustainability and financial stability.
- PCL and the paper industry faces notable environmental sustainability challenges due to deforestation, waste generation, pollution, invasive species, and chemical exposure.
- PCL's economic sustainability can be enhanced by investments in technology and processes that reduce environmental impact.
- Maintaining compliance with international financial reporting standards are crucial not only for economic sustainability but also for maintaining trust with stakeholders.

9.6. Recommendation

- Address concerns related to child labor, worker exploitation, and chemical exposure to ensure the safety and well-being of its workforce.
- Increase efforts to mitigate the negative impacts of its operations like deforestation and pollution.
- Consider alternative, sustainable sources of raw materials to reduce the environmental impact and improve social sustainability.
- Develop and adopt cleaner production processes to mitigate pollution and waste generation.
- Carefully assess the introduction of non-native tree species to avoid potential environmental harm.
- Emphasize transparency and accountability, not only in financial reporting but also in addressing sustainability concerns.
- Consider seeking third-party certifications related to sustainability to build trust and credibility with consumers and investors.

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10. Appendices:

10.1. Appendix 1 - Analysis of Adjusted Statement of Profit & Loss:

	30-06-2023 (Reported)	Adjustments	30-06-2023 (Adjusted)	30-Jun-22	Change	Change %
Revenue, net	14,609,357,421	-	14,609,357,421	12,067,714,190	2,541,643,231	21.1%
Cost of sales	(11,683,899,628)	3,500,000	(11,680,399,628)	(9,568,034,401)	(2,112,365,227)	22.1%
Gross profit	**2,925,457,793**	**3,500,000**	**2,928,957,793**	**2,499,679,789**	**429,278,004**	**17.2%**
Administrative expenses	(456,837,667)	(2,699,876)	(459,537,543)	(478,618,630)	19,081,087	-4.0%
Selling & distribution expenses	(258,744,617)	-	(258,744,617)	(256,098,753)	(2,645,864)	1.0%
Finance costs	(1,296,005,232)	(214,156,793)	(1,510,162,025)	(1,211,931,443)	(298,230,582)	24.6%
Profit before distribution of WPP & WF	**913,870,277**	**(213,356,669)**	**700,513,608**	**553,030,963**	**147,482,645**	**26.7%**
Workers' profit participation & welfare fund	(45,693,514)		(45,693,514)	(27,651,549)	(18,041,965)	65.2%
Profit before tax	**868,176,763**	**(213,356,669)**	**654,820,094**	**525,379,414**	**129,440,680**	**24.6%**
Current tax (expense)/income	(172,775,897)	-	(172,775,897)	2,075,508	(174,851,405)	-8424.5%
Deferred tax (expense)/income	(35,198,285)	-	(35,198,285)	7,092,215	(42,290,500)	-596.3%
Profit after tax	**660,202,581**	**(213,356,669)**	**446,845,912**	**534,547,137**	**(87,701,225)**	**-16.4%**

10.2. Appendix 2 - Analysis of Adjusted Balance Sheet:

	30-06-2023 (Reported)	Adjustments	30-06-2023 (Adjusted)	30-Jun-22	Change	Change %
Assets						
Non-current assets						
Property, plant and equipment, net	29,933,268,310	(4,500,000)	29,928,768,310	22,354,998,136	7,573,770,174	34%
Intangible Assets, net	203,132,105	-	203,132,105	196,028,113	7,103,992	4%
Right-of-use assets	19,881,667	-	19,881,667	35,316,991	(15,435,324)	-44%
Capital work-in-progress	821,806,406	(165,206,669)	656,599,737	1,124,398,926	(467,799,189)	-42%
Investment	344,500,000	-	344,500,000	344,500,000	-	0%
	31,322,588,488	**(169,706,669)**	**31,152,881,819**	**24,055,242,166**	**7,097,639,653**	**30%**
Current assets						
Inventories	10,750,871,244	3,500,000	10,754,371,244	10,224,836,983	529,534,261	5%
Trade and other receivables	1,319,039,957	-	1,319,039,957	996,866,933	322,173,024	32%
Advances, deposits and prepayments	1,381,025,341	-	1,381,025,341	1,542,189,864	(161,164,523)	-10%
Cash and bank equivalents	536,818,927	(17,800,000)	519,018,927	541,317,599	(22,298,672)	-4%
	13,987,755,469	**(14,300,000)**	**13,973,455,469**	**13,305,211,379**	**668,244,090**	**5%**
Total assets	**45,310,343,957**	**(184,006,669)**	**45,126,337,288**	**37,360,453,545**	**7,765,883,743**	**21%**
EQUITY AND LIABILITIES						
Capital and reserves						
Share capital	4,520,747,010	-	4,520,747,010	4,520,747,010	-	0%
Revaluation reserve	8,709,246,925	-	8,709,246,925	2,998,673,400	5,710,573,525	190%
Retained earnings	3,761,579,721	(213,356,669)	3,548,223,052	3,355,024,616	193,198,436	6%
	16,991,573,656	**(213,356,669)**	**16,778,216,987**	**10,874,445,026**	**5,903,771,961**	**54%**
Non-current Liability						
Long term borrowings	17,387,736,934	-	17,387,736,934	16,996,071,544	391,665,390	2%
Lease Liabilities	-	-	-	269,871,765	(269,871,765)	-100%
Deferred tax liability	1,422,507,425	-	1,422,507,425	1,210,153,243	212,354,182	18%
	18,810,244,359	**-**	**18,810,244,359**	**18,476,096,552**	**334,147,807**	**2%**
Current liabilities						
Long term borrowings-current portion	2,596,577,205	-	2,596,577,205	2,803,931,499	(207,354,294)	-7%
Short term borrowings	5,191,758,021	-	5,191,758,021	3,425,208,583	1,766,549,438	52%
Trade and other payables	1,558,427,347	20,450,000	1,578,877,347	1,488,323,381	90,553,966	6%
Provision for expenses	1,510,435	8,900,000	10,410,435	67,266,891	(56,856,456)	-85%
Income tax provision	160,252,934	-	160,252,934	225,181,613	(64,928,679)	-29%
	9,508,525,942	**29,350,000**	**9,537,875,942**	**8,009,911,967**	**1,527,963,975**	**19%**
Total liabilities	**28,318,770,301**	**29,350,000**	**28,348,120,301**	**26,486,008,519**	**1,862,111,782**	**7%**
Total equity and liabilities	**45,310,343,957**	**(184,006,669)**	**45,126,337,288**	**37,360,453,545**	**7,765,883,743**	**21%**

10.3. Appendix 3 - Mix analysis:

			Contribution	
	30-Jun-23	**30-Jun-22**	**30-Jun-23**	**30-Jun-22**
Revenue				
Paper product	6,136,238,285	4,585,731,392	42%	38%
Tissue & Hygine product	8,473,119,136	7,481,982,798	58%	62%
	14,609,357,421	**12,067,714,190**	100%	100%
Revenue by source				
Local sale	13,052,736,365	10,752,092,560	89%	89%
Export sale	1,556,621,056	1,315,621,630	11%	11%
	14,609,357,421	**12,067,714,190**	100%	100%
Cost of Revenue				
Paper product	4,673,559,852	3,348,812,040	40%	35%
Tissue & Hygine product	7,010,339,777	6,219,222,361	60%	65%
	11,683,899,629	**9,568,034,401**	100%	100%
GP- Paper Product				
Revenue- Paper product	6,136,238,285	4,585,731,392		
COS- Paper product	(4,673,559,852)	(3,348,812,040)		
	1,462,678,433	**1,236,919,352**		
GP%- Paper Product	**23.8%**	**27.0%**		
GP- Tissue & Hygine product				
Rev-Tissue & Hygine product	8,473,119,136	7,481,982,798		
COS-Tissue & Hygine product	(7,010,339,777)	(6,219,222,361)		
	1,462,779,359	**1,262,760,437**		
GP%- Tissue & Hygine product	**17.3%**	**16.9%**		

10.4. Appendix 4: Ratio Analysis:

SL.	Ratio Name	30-Jun-23	30-Jun-22
1	Current Ratio	1.5	1.7
2	Quick Ratio	0.3	0.4
3	Debt to equity ratio	1.69	2.44
4	Asset Turnover Ratio	0.32	0.32
5	TIE Ratio	1.4	1.4
6	Return on Asset	1.0%	1.4%
7	Return on Equity	2.7%	4.9%

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